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Task: Pegs and Disks problem

1. Problem Description

The Pegs and Disks problem is a classical Artificial Intelligence problem known as the Tower of Hanoi. It consists of three pegs, named A, B, and C, and three disks of different sizes. Initially, all three disks are placed on peg A, arranged from largest at the bottom to smallest at the top.

The objective is to move all three disks from peg A to peg B, which is the goal state, using peg C as an auxiliary peg.

The problem is solved by applying operators that move the topmost disk from one peg to the top of another peg.

2. Initial State and Goal State

- Initial State: All three disks are on peg A, with the largest at the bottom and the smallest at the top.
- Goal State: All three disks are on peg B in the same size order, with the largest at the bottom and the smallest at the top.
- Available Pegs: A, B, and C.
- Number of Disks: 3.

3. Operators (Rules)

The following rules must be followed while solving the problem:

1. Only one disk can be moved at a time.

2. Only the topmost disk on a peg can be moved.
3. A disk can be moved to an empty peg or onto a larger disk.
4. A larger disk cannot be placed on a smaller disk.
5. The objective is to transfer all disks from peg A to peg B using peg C as temporary storage.

These rules define the legal actions available to the problem-solving agent. **4. Step-by-Step Solution** Let:

- D1 = Smallest disk
- D2 = Medium disk
- D3 = Largest disk

To move all three disks from peg A to peg B, the following sequence of legal moves is used.

Step	**Disk Moved**	**Movement**	**Description**
1	D1 (smallest)	A → B	Move the smallest disk from A to B.
2	D2 (medium)	A → C	Move the medium disk from A to C.
3	D1 (smallest)	B → C	Place the smallest disk on top of the medium disk at C.
4	D3 (largest)	A → B	Move the largest disk from A to B.
5	D1 (smallest)	C → A	Move the smallest disk from C back to A.
6	D2 (medium)	C → B	Move the medium disk from C onto the largest disk at B.
7	D1 (smallest)	A → B	Move the smallest disk from A onto the medium disk at B.

5. State Representation

The state of the problem can be represented by showing the disks on each peg after every move. The order of disks in each set is from bottom to top.

State	Peg A	Peg B	Peg C
Initial	D3, D2, D1	Empty	Empty
After Step 1	D3, D2	D1	Empty
After Step 2	D3	D1	D2
After Step 3	D3	Empty	D2, D1
After Step 4	Empty	D3	D2, D1
After Step 5	D1	D3	D2
After Step 6	D1	D3, D2	Empty
Goal State	Empty	D3, D2, D1	Empty

In the goal state, all three disks are on peg B in the correct order, with the largest disk at the bottom and the smallest disk at the top.

6. Minimum Number of Moves

The minimum number of moves required to transfer \(n\) disks in the Tower of Hanoi problem is:

$$T(n)=2^n-1$$

For three disks:

$$T(3)=2^3-1=8-1=\boxed{7}$$

Therefore, the minimum number of moves required is 7.

7. Conclusion

The Pegs and Disks problem demonstrates how an AI agent can solve a problem by representing states, defining legal operators, and searching for a sequence of actions that reaches the goal state. By following the rules and moving one disk at a time, all three disks can be transferred from peg A to peg B in the correct order.

Final Answer: The goal state is achieved in 7 moves, with all three disks placed on peg B and pegs A and C left empty.